

MATLAB Simulation Codes for [3, Sect. 4]

This MATLAB code reproduces the simulation results from [3, Sect. 4].

Description

The considered class of $n + m$ hyperbolic partial differential equations (PDEs) is of the form

$$\mathbf{u}_t(t, x) + \mathbf{\Lambda}_+(x)\mathbf{u}_x(t, x) = \frac{1}{n}\mathbf{\Sigma}(x)\mathbf{u}(t, x) + \mathbf{W}(x)\mathbf{v}(t, x), \quad (1a)$$

$$\mathbf{v}_t(t, x) - \mathbf{\Lambda}_-(x)\mathbf{v}_x(t, x) = \frac{1}{n}\mathbf{\Theta}(x)\mathbf{u}(t, x) + \mathbf{\Psi}(x)\mathbf{v}(t, x), \quad (1b)$$

with boundary conditions

$$\mathbf{u}(t, 0) = \mathbf{Q}\mathbf{v}(t, 0), \quad \mathbf{v}(t, 1) = \mathbf{R}\mathbf{u}(t, 1) + \mathbf{U}(t), \quad (2)$$

where $\mathbf{u} = (u^i)_{i=1}^n$, $\mathbf{v} = (v^j)_{j=1}^m$ are the PDE states and $\mathbf{U} = (U^j)_{j=1}^m$ is the control input. In the framework of [3, Sect. 4], the parameters of the system (1), (2) are given by

$$\mathbf{\Lambda}_+(x) = I_{n \times n}, \quad \mathbf{\Lambda}_- = \text{diag}(2, 1), \quad (3a)$$

$$\sigma_{i,\ell}(x) = x^3(x+1) \left(\frac{i}{n} - \frac{1}{2} \right) \frac{\ell}{n} \left(\frac{\ell}{n} - 1 \right), \quad (3b)$$

$$W_{i,1}(x) = 3 \left(\frac{i}{n} - \frac{1}{2} \right), \quad W_{i,2}(x) = 2 \left(\frac{i}{n} - \frac{1}{2} \right), \quad (3c)$$

$$\theta_{1,i}(x) = -\frac{3i}{n} \left(\frac{i}{n} - 1 \right), \quad \theta_{2,i}(x) = -\frac{2i}{n} \left(\frac{i}{n} - 1 \right), \quad (3d)$$

$$\psi_{l,j}(x) = 0, \quad l, j \in \{1, 2\}, \quad (3e)$$

$$Q_{i,1} = 8 \left(\frac{i}{n} - \frac{1}{2} \right), \quad Q_{i,2} = -8 \left(\frac{i}{n} - 2 \right), \quad (3f)$$

$$R_{1,i} = \cos \left(2\pi \frac{i}{n} \right), \quad R_{2,i} = 2 \frac{i}{n} \left(\frac{i}{n} + 5 \right), \quad (3g)$$

for $i, \ell = 1, \dots, n$ and $x \in [0, 1]$, where n can be chosen freely and $m = 2$.

A respective $\infty + m$ continuum system of the form

$$u_t(t, x, y) + \lambda(x, y)u_x(t, x, y) = \int_0^1 \sigma(x, y, \eta)u(t, x, \eta)d\eta + \mathbf{W}(x, y)\mathbf{v}^c(t, x), \quad (4a)$$

$$\mathbf{v}_t^c(t, x) - \mathbf{\Lambda}_-(x)\mathbf{v}_x^c(t, x) = \int_0^1 \mathbf{\Theta}(x, y)u(t, x, y)dy + \mathbf{\Psi}(x)\mathbf{v}^c(t, x), \quad (4b)$$

with boundary conditions

$$u(t, 0, y) = \mathbf{Q}(y)\mathbf{v}^c(t, 0), \quad \mathbf{v}^c(t, 1) = \int_0^1 \mathbf{R}(y)u(t, 1, y) + \mathbf{U}(t), \quad (5)$$

is constructed to approximate the original $n + m$ system (1), (2), where the continuum parameters corresponding to (3) can be chosen as

$$\lambda(x, y) = 1, \quad \mathbf{\Lambda}_- = \text{diag}(2, 1), \quad (6a)$$

$$\sigma(x, y, \eta) = x^3(x + 1) \left(y - \frac{1}{2} \right) \eta (\eta - 1), \quad (6b)$$

$$W_1(x, y) = 3 \left(y - \frac{1}{2} \right), \quad W_2(x, y) = 2 \left(y - \frac{1}{2} \right), \quad (6c)$$

$$\theta_1(x, y) = -3y(y - 1), \quad \theta_2(x, y) = -2y(y - 1), \quad (6d)$$

$$\psi_{i,j}(x) = 0, \quad i, j \in \{1, 2\}, \quad (6e)$$

$$Q_1(y) = 8 \left(y - \frac{1}{2} \right), \quad Q_2(y) = -8(y - 2), \quad (6f)$$

$$R_1(y) = \cos(2\pi y), \quad R_2(y) = 2y(y + 5). \quad (6g)$$

i.e., by replacing i/n and l/n with y and η , respectively, in (3). More precisely, a continuum observer is constructed to approximately estimate the state of the $n + m$ system based on the measurement $\mathbf{v}(t, 0)$. The continuum observer is of the form

$$\begin{aligned} \hat{u}_t(t, x, y) + \lambda(x, y)\hat{u}_x(t, x, y) + \mathbf{P}_+(x, y)(\hat{\mathbf{v}}(t, 0) - \mathbf{v}(t, 0)) = \\ \int_0^1 \sigma(x, y, \eta)\hat{u}(t, x, \eta)d\eta + \mathbf{W}(x, y)\hat{\mathbf{v}}(t, x), \end{aligned} \quad (7a)$$

$$\begin{aligned} \hat{\mathbf{v}}_t(t, x) - \mathbf{\Lambda}_-(x)\hat{\mathbf{v}}_x(t, x) + \mathbf{P}_-(x)(\hat{\mathbf{v}}(t, 0) - \mathbf{v}(t, 0)) = \\ \int_0^1 \mathbf{\Theta}(x, y)\hat{u}(t, x, y)dy + \mathbf{\Psi}(x)\hat{\mathbf{v}}(t, x), \end{aligned} \quad (7b)$$

with boundary conditions

$$\hat{u}(t, 0, y) = \mathbf{Q}(y)\mathbf{v}(t, 0), \quad \hat{\mathbf{v}}(t, 1) = \int_0^1 \mathbf{R}(y)\hat{u}(t, 1, y)dy + \mathbf{U}(t), \quad (8)$$

where $\mathbf{P}_+, \mathbf{P}_-$ are given by

$$\mathbf{P}_+(x, y) = \mathbf{M}(x, 0, y)\mathbf{\Lambda}_-(0), \quad (9a)$$

$$\mathbf{P}_-(x) = \mathbf{N}(x, 0)\mathbf{\Lambda}_-(0), \quad (9b)$$

where $\mathbf{M}, \mathbf{N}$ are the backstepping continuum observer kernels satisfying [1, (11)–(13)]. For the

parameters (3), the kernels $\mathbf{M}, \mathbf{N}$ are given in closed-form by [1, (23)]

$$M_1^1(x, \xi, y) = \left(y - \frac{1}{2}\right), \quad (10a)$$

$$M_1^2(x, \xi, y) = e^{x-\frac{1}{2}\xi-1} \left(y - \frac{1}{2}\right), \quad (10b)$$

$$M_2^2(x, \xi, y) = e^{x-\xi} \left(y - \frac{1}{2}\right), \quad (10c)$$

$$N_{1,1}^1(x, \xi) = 0, \quad N_{1,1}^2(x, \xi) = 0, \quad (10d)$$

$$N_{2,1}^1(x, \xi) = 0, \quad N_{2,1}^2(x, \xi) = e^{x-\frac{1}{2}\xi-1}, \quad (10e)$$

$$N_{1,2}^2(x, \xi) = 0, \quad N_{2,2}^2(x, \xi) = e^{x-\xi}, \quad (10f)$$

where $M_1^\star(\cdot, y)$, $N_{1,1}^\star$, and $N_{2,1}^\star$ are defined on $\mathcal{T}_1^1 = \{(x, \xi) \in \mathcal{T} : \xi \geq 2x - 1\}$ and $\mathcal{T}_1^2 = \{(x, \xi) \in \mathcal{T} : \xi \leq 2x - 1\}$ for the respective superindex $\star = 1, 2$, while $M_2^2(\cdot, y)$, $N_{2,1}^2$ and $N_{2,2}^2$ are defined on $\mathcal{T}_2^2 = \mathcal{T}$, for each $y \in [0, 1]$.

Finally, the continuum-observer based control law is of the form

$$\mathbf{U}(t) = \int_0^1 \int_0^1 \mathbf{K}(1, \xi, y) \hat{u}(t, \xi, y) dy d\xi + \int_0^1 \mathbf{L}(1, \xi) \hat{\mathbf{v}}(t, \xi) d\xi - \int_0^1 \mathbf{R}(y) \hat{u}(t, 1, y) dy, \quad (11)$$

where $\mathbf{K}, \mathbf{L}$ are the backstepping continuum control kernels satisfying [2, (12)–(16)]. For the parameters (3), the kernels $\mathbf{K}, \mathbf{L}$ are given in closed-form by [2, (75)]

$$K_1^1(x, \xi, y) = y(y - 1), \quad (12a)$$

$$K_1^2(x, \xi, y) = e^{x-2\xi} y(y - 1), \quad (12b)$$

$$K_2^2(x, \xi, y) = e^{2(x-\xi)} y(y - 1), \quad (12c)$$

$$L_{1,1}^1(x, \xi) = L_{1,1}^2(x, \xi) = 0, \quad (12d)$$

$$L_{1,2}^1(x, \xi) = 0, \quad L_{1,2}^2(x, \xi) = -2e^{x-2\xi}, \quad (12e)$$

$$L_{2,1}^2(x, \xi) = 0, \quad L_{2,2}^2(x, \xi) = -2e^{2(x-\xi)}, \quad (12f)$$

where $K_1^\star(\cdot, y)$, $L_{1,1}^\star$, and $L_{1,2}^\star$ are defined on $\mathcal{T}_1^1 = \{(x, \xi) \in [0, 1]^2 : \frac{1}{2}x \leq \xi \leq x\}$ and $\mathcal{T}_1^2 = \{(x, \xi) \in [0, 1]^2 : \xi \leq \frac{1}{2}x\}$ for the respective superindex $\star = 1, 2$, while $K_2^2(\cdot, y)$, $L_{2,1}^2$ and $L_{2,2}^2$ are defined on $\mathcal{T}_2^2 = \mathcal{T} = \{(x, \xi) \in [0, 1]^2 : \xi \leq x\}$, for each $y \in [0, 1]$.

Running the MATLAB script

`n2contobsv.m`

reproduces the simulation results from [3, Sect. 4]. The script simulates a finite-difference approximation (with 128 grid points) of the $n + 2$ system (1), (2) with parameters (3) for $n = 53, 55, 57, 59$ in closed-loop with the continuum observer-based control law (11), (12), where the continuum observer (7)–(10) is implemented as a $60 + 2$ system.

Examples

The simulation results for [3, Sect. 4] can be reproduced by running the script `n2contobsv.m`.

References

- [1] J.-P. Humaloja and N. Bekiaris-Liberis. Observer-based output-feedback stabilization of continua of linear hyperbolic PDEs using backstepping. *IFAC-PapersOnLine*, 59:143–148, 2025.
- [2] J.-P. Humaloja and N. Bekiaris-Liberis. Backstepping control of continua of linear hyperbolic PDEs and application to stabilization of large-scale $n + m$ coupled hyperbolic PDE systems. *Automatica*, 183:112647, 2026.
- [3] J.-P. Humaloja and N. Bekiaris-Liberis. Output-feedback backstepping stabilization of large-scale $n + m$ coupled hyperbolic PDEs via continuum kernels and observer. *Automatica*, 2026 (to appear).